Akanksha Sachan

Contact

PhD Candidate in Computational Biology
Joint Carnegie Mellon–University of Pittsburgh PhD Program
Machine Learning for Epigenetic reprogramming
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Education

University of Pittsburgh School of Medicine — *'24 – Present*
PhD Candidate in Computational Biology — *Expected '27*
Advised by Dr. Jishnu Das

CMU, Kenneth C. Griffin School of Computer Science — *'22 – '24*
PhD Student in Computational Biology
Computational Biology Department | GPA: 3.58/4

Indian Institute of Technology (IIT), Bombay — *'18 – '22*
BTech. in Chemical Engineering | GPA: 8.06/10

Awards and Honors

American Federation for Aging Research (AFAR) Graduate Scholarship — *'25*
Best Poster Award at the Aging Institute Research Day, UPitt — *'24*
1st Place, Student A100 track | NeurIPS LLM Efficiency Challenge | team of 5 — *'23*
1st Place, Accessibility Track | Pitt Challenge Hackathon | team of 3 — *'22*
Recipient of **Undergraduate Research Award** (URA-01) by IIT Bombay — *'22*
Recipient of Merit-cum-Means (MCM) Scholarship by IIT Bombay — *'21*
Top 0.2% (Rank 3188 out of **1.1M+** candidates) in JEE (Main) — *'18*
Top 1.5% (Rank 2517 out of **160k+** candidates) in JEE (Advanced) — *'18*

Invited Talks

"Interpretable machine learning coupled to gene regulatory networks uncovers subcircuits underlying cell fate decisions" | *preprint*

National Institutes of Health (NIH) IGVF Consortium Annual Meeting — *Sep '26*
International Society for Computational Biology (ISCB)/(ISMB) — *Jul '26, '25*
Keystone Symposia: Artificial Intelligence in Molecular Biology — *Sep '25*
American Association of Immunologists (AAI) Annual Meeting 2026 — *Apr '26*
Center of Lung Aging and Regeneration, School of Medicine, UPitt — *Nov '25*
30th Annual School of Medicine Graduate Student Research Symposium, UPitt — *Oct '25*

"Sex-dimorphic DNA lesion accumulation drives a PDK4 metabolic checkpoint in skeletal muscle" | *in-review at Nature Aging*

Center for Systems Immunology Annual Retreat, UPitt — *Nov '25*
23rd Annual Department of Immunology Scientific Retreat, UPitt — *Sep '25*
Aging Institute Research Day, UPitt — *Nov '24*

"Micro-MPA digital twin, with sclerostin-dependent RANKL production reproduces pivotal dual-action effect of romosozumab found in clinical trials"

50th European Calcified Tissue Society (ECTS) Congress — *Apr '23*

Research Experience Prior to Thesis

Mapping the epigenetic landscape of single-cells using 3-D chromatin architecture *'22 – '24*

Supervisor – Prof. Jian Ma, Computational Biology Department, SCS, CMU

Developed a framework to identify cis-regulatory 3-D genomic hubs controlling target-gene expression and key transcription factors in healthy and cancerous mouse-brain single-cell, multi-omic profiling.

Translational Mechanobiology Lab, MBI, National University of Singapore *'22*

Summer Internship | Supervisor – Prof. Hanry Yu, Dept. of Physiology, YLL School of Medicine

Built a Generative-Adversarial-Network (GAN) pipeline for colorectal-cancer imaging to support clinical deep-learning deployment

Lab for Bone Biomechanics, Institute of Biomechanics, ETH Zurich *'21*

Summer Internship | Supervisor – Prof. Ralph Muller, Dept. of Health Sci&Tech

Added a Type-2 Diabetes pathway to an agent-based osteoporosis model and ran large-scale simulations on the CSCS Swiss National Supercomputing Center

Multicellular Systems Engineering Lab, University of Notre Dame *'20*

Summer Internship | Supervisor – Prof. Jeremiah Zartman, Chemical & Biomolecular Engineering

Built a morphometric feature-extraction pipeline for *Drosophila* wing imaginal discs (OpenCV, Fiji) to study organism-wide cell-signaling

Key Coursework

Machine Learning & Statistics: Intermediary Statistics (CMU: 36-705), Probabilistic Graphical Models (CMU: 10-708), Introduction to Machine Learning (CMU: 10-701)

Predicting genomic-hub mediated TF-perturbation response via geometric deep learning *'23*

10-708: Probabilistic Graphical Models

Augmented a GNN for Perturb-seq response prediction with a Hi-C contact graph (Cancer cell-line), adding 3D-genome topology to the co-expression and Gene Ontology graph features.

Single-cell DNA methylation prediction using single-cell HiC data and Graph NNs *'23*

02-710: Computational Genomics

Developed a **GNN** to predict methylation from HiC data, connecting structure to phenotype; clustered **single-nucleus DNA methylation** data with K-means and t-SNE.

Athletic Accolades

Bronze Medal | 35th Intercollegiate (inter-IIT) Aquatics Meet | 3rd in Medley-Relay *'19*

3rd Position | Comp-Bio Department, Pretty Good Race (5km) for SCS, CMU *'23*

Swam 12-hours | Swimathon organized by the IIT Bombay aquatics committee *'19*

Ran 26.2 miles | Dick's Sporting Goods Pittsburgh Marathon | 5 hours 15 mins *'24*

Ran 13.1 miles | Dick's Sporting Goods Pittsburgh Half Marathon | 2 hours 30 mins *'23*

Leadership & Societal Contributions

President *Jun '26 – Present*
Allegheny Science Policy and Governance (ASPG)
Previously Vice President (Jul '25 – Jun '26). Bridging the gap between **scientists and Pennsylvania legislators** through interactions at the **State Capitol**.

Technights Coordinator *Aug '23 – Aug '25*
Carnegie Mellon University, Robotics Institute
Organized and led a **computer-science outreach** program teaching middle-school girls sessions on foundation models, and deep learning.

NIH Consortium Hackathon, Co-organizer & Team Lead *Mar '24*
4D Nucleome, Seattle, Washington
Led a team of 5 to build integrative pipelines for single-cell HiC analysis from cell-atlas-scale datasets

Reviewer
ICML Foundation Model for Life Sciences Workshop 2026, *PNAS Journal* (*assisted PI*), *Aging Cell Journal* (*assisted PI*)

Teaching Assistant *Fall '23 – Spring '24*
02-760: Lab Methods for Computational Biologists, SCS, CMU
Taught a lecture on **genomics technologies** to a class of ∼20 students, revamping the RNA sequencing module of the course.

Treasurer • Senator *Aug '23 – Aug '25*
Graduate Student Association (*Joint CMU-UPitt Comp-Bio PhD Program*)

Editorial Board Member • Design Convenor *'19 – '21*
Insight, the student newsletter of IIT Bombay

Convenor, Girls Swim Team *'19 – '20*
Aquatics, Indian Institute of Technology Bombay

Consortiums & Workshops

NIH Consortium Annual Meetings *'23, '25*
IGVF, SenNet:Cellular Senescence Network, 4D Nucleome
Grace Hopper Celebration *Oct '24*
AnitaB.org, Philadelphia
Computational Genomics Summer Institute *Jul '23*
University of California, Los Angeles
Grad Cohort for Women *April '23*
Computing Research Association (*CRA-WP*), *San Francisco*

Compute Allocation Grants

ACCESS (NSF Advanced Cyberinfrastructure Coordination Ecosystem: Services & Support) **1.5M** credits on the NCSA Delta GPU supercomputer *'26*